

Sayantan Paul

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Overview

- Experience working with **Python, NumPy, pandas, scikit-learn, PyTorch, TensorFlow.**
- Familiar with AWS core services including **Amazon S3, RDS, DynamoDB, Redshift, Glue, Lambda, Athena, Step Functions, CloudWatch, CloudTrail, IAM, SQS.**
- Proficient with **SQL - schema design, joins, indexing, query optimization, stored procedures, transaction management.**
- Skilled in **Windows Server Administration, Kubernetes, Group Policy, Active Directory, VMWare, Citrix, Incident Management.**

Technical Skills

- **Data & ML:** Pandas, NumPy, Scikit-learn, TensorFlow, Keras, Random Forest, Neural Networks
- **Languages & Frameworks:** Python, JavaScript, React.js, Django, HTML5, CSS3, C++
- **Cloud & DevOps:** AWS EC2, S3, Lambda, API Gateway, CloudWatch, IAM, CloudFormation
- **Databases:** MySQL, AWS RDS, Redshift, PostgreSQL
- **Tools:** VS Code, Jupyter Notebook, Git, GitHub, Docker, Postman

Education

Texas A&M University August 2025
Masters of Science in Computer Science. GPA 4.0/4.0.

Experience

Texas A&M University **Kingsville, TX**
Graduate Research Assistant May 2024 - August 2025

- Designed **deep learning classifiers** with **embedding layers, SeparableConv / 1D Conv, batch normalization, and dense layers.**
- Implemented **Class Activation Maps (CAMs)** for model interpretability and **1D heatmaps.**
- Benchmarked with statistical metrics (**Manhattan, Euclidean, Cosine**) and evaluated with **accuracy, recall, specificity.**
- Applied Generative AI to biological data with a **deep learning pipeline, pre-trained Hugging Face model, and GitHub code.**
- Built modular **Python pipelines** with **OOP, preprocessing, augmentation, and data analysis** (NumPy arrays, dictionaries, histograms).
- Trained **TensorFlow/Keras VAEs** with **convolution, masking, sampling layers, leveraging pre-trained embeddings and normalization.**
- Designed training workflows with **Adam optimizer, early stopping, CSV logging, and hyperparameter tuning.**
- Published a **paper** at **IEEE IMSA Conference, Cairo, Egypt** (July 2025).

Capgemini **India**
Senior Analyst April 2018 - March 2022

- Accomplished **client-server hardening** against **ransomware attacks.**
- Maintenance for **on-premises** and **Cloud-based (Microsoft Azure)** servers.
- Managed **MySQL database systems** and **BI reporting** for **HR, Payroll, and Finance** supporting **90K+** retail employees.